MAcRIO

Cross-Border Food Ingredient Scanner & Utensil Toxicity Platform

Target Platform: IBM LinuxONE z16 (s390x Architecture)
Hardware AI Acceleration: IBM Telum On-Chip NNPA via zDNN
AI Intelligence: Autonomous Agentic Google Gemma LLM

Team: TunakTunakTun
Event: IBM Z & LinuxONE Datathon 2026
Deliverables: Full Stack Web, Native Android APK, Mainframe API

Exposing predatory cross-border double standards, deciphering misleading marketing, and providing personalized clinical safety guardrails for chronic conditions and allergies.

1. The Problem: Food Asymmetry & Double Standards

Multinational food brands exploit weak regional regulations while consumers face chronic health crises.

■ Cross-Border Double Standards Global FMCG conglomerates reformulate products sold in India & SE Asia with cheap palm oil, maltodextrin, high fructose corn syrup, and synthetic food dyes (E171, E102) while selling clean versions in the EU and US.	**■ Rising Chronic Diseases & Allergies** Diabetes, hypertension, and severe food allergies (peanuts, gluten, dairy) are surging. Shoppers cannot decipher complex chemical additives or assess whether a product contraindicates their personal health status.
■■ Deceptive Front-of-Pack Marketing Products labeled 'High Protein', 'Whole Wheat', or 'Zero Sugar' often hide excessive sodium, disguised sugars, and ultra-processed additives that evade casual consumer detection.	**■ Cookware & Microplastic Leaching** Degraded, scratched PTFE non-stick pans (Teflon) and single-use microplastic-leaching containers (RIC 3, 6, 7) release toxic forever chemicals (PFAS) directly into daily home-cooked meals.

2. The Solution: Autonomous Agentic Intelligence

Instant, multi-turn clinical evaluation powered by Google Gemma on IBM LinuxONE.

■ Autonomous Agentic Search Loop	■■ Singapore Nutri-Grade (A to D)	■ Food Pharmer Cleaner Swaps
Gemma does not rely on a single static database. It acts as an autonomous clinical agent: issues live queries via Google Serper, audits missing ingredients, calls follow-up searches, and synthesizes complete nutrition profiles.	Implements Singapore's Health Promotion Board (HPB) mandatory Nutri-Grade scoring (A, B, C, D) based on sugar and saturated fat thresholds per 100g/ml across both packaged drinks and street foods.	Recommends actionable, healthier alternatives inspired by Revant Himatsingka's 'Label Padhega India' framework (e.g., swapping refined palm oil snacks for roasted seed alternatives).
■■ Personalized Contraindication Audit	**■ Utensil & Cookware Safety**	**■ Privacy-Preserving Mainframe Core**
Instantly cross-references patient allergies (e.g., Tree nuts) and chronic illnesses (Hypertension, Diabetes) against extracted ingredients to issue immediate FIT or AVOID warnings.	Computer vision and material classification engine that identifies Resin Identification Codes (RIC 1-7) and flags toxicity risks from scratched non-stick pans and low-grade plastics.	All sensitive patient health telemetry and barcode scanning streams are processed on IBM LinuxONE in tamper-proof enterprise confidential computing enclaves.

3. System Architecture on IBM LinuxONE

Enterprise-grade pipeline integrating hardware coprocessors, LLM inference, and global databases.

Layer	Technology / Component	Role in MAcRIO Platform
Hardware & OS	IBM LinuxONE z16 (s390x architecture) Ubuntu 24.04 LTS	Provides 99.999% mainframe uptime, massive memory bandwidth, and enterprise-grade hardware cryptography.
AI Hardware Acceleration	IBM Telum On-Chip AI Coprocessor (NNPA) IBM zDNN v1.1.2 + FlashAttention	Offloads matrix multiplication and attention tensor layers directly to on-die NNPA execution units, accelerating Gemma inference without GPUs.
Core Application	FastAPI (Python 3.12 s390x native) RealDictCursor connection pool	High-concurrency async REST API serving barcode lookups, personalized clinical scans, and camera OCR processing.
Data Storage	PostgreSQL on LinuxONE (TLSv1.3 encrypted) Offline s390x SnapML pipelines	Houses super-datasets: Open Food Facts, ICMR-NIN Indian Food Composition Tables (IFCT), USDA, and Singapore HPB datasets.
Client Applications	Native Android APK (v1.0.0) Responsive Progressive Web App (PWA)	Live barcode scanner, camera lens OCR menu rater, patient health profile manager, and utensil toxicity guide.

4. Proof of Execution on IBM LinuxONE z16

Live terminal execution logs captured directly from our s390x mainframe instance.

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1. Mainframe Kernel & Architecture (uname -a):
Linux tunaktunaktun-macrio-ibm-z-datathon 6.8.0-147-generic #147-Ubuntu SMP s390x GNU/Linux

2. IBM S/390 CPU Architecture & Machine Type 9175 (lscpu):
Architecture: s390x | CPU op-mode(s): 64-bit | Vendor ID: IBM/S390 | Machine type: 9175

3. IBM Telum NNPA Hardware Device Detection (llama-server --list-devices):
Available devices:
zDNN: IBM Z Neural Network Processing Assist (NNPA) (0 MiB, 0 MiB free)

4. Live Health Check Endpoint Response (curl -s http://127.0.0.1:8000/health):
{ "status": "online", "platform": "IBM LinuxONE (s390x)", "processor_acceleration": "IBM Telum On-Chip AI Assisted", "database": {"name":
"macrio_db (PostgreSQL TLSv1.3)", "singapore_hawker_dishes": 10, "fast_food_items": 5} }
```

5. Autonomous Multi-Search Agent in Action

Step-by-step trace showing Gemma autonomously discovering, verifying, and scoring a product.

Stage	Gemma Autonomous Action	Hardware / Engine Execution
Turn 1: Barcode SKU Search	Issues `ACTION: SEARCH 8908017270008`. Google Serper discovers product SKU: 'Nutpac pistachio 250g - Gulabchand's Ecomart'.	Prompt evaluated on IBM Telum NNPA via zDNN. FlashAttention accelerates initial token ingestion.
Turn 2: Information Gap Analysis	Gemma identifies missing ingredients & nutritional values. Autonomously issues: `ACTION: SEARCH Nutpac pistachio 250g ingredients nutrition`. Retreives exact ingredients: 'Pistachio, Edible Salt'.	KV Cache prefix reuse (LCP similarity = 0.76) slashes prompt evaluation time from 48s to 6s.
Turn 3: Clinical Synthesis & Scoring	Cross-references patient profile (`allergies: ['Tree nuts']`, `conditions: ['Hypertension']`). Flags `fit_verdict: 'AVOID'` due to tree nut conflict and flags Edible Salt for hypertension. Assigns Nutri-Grade B and recommends raw unsalted swap.	Structured JSON validated against clinical schema. Stored in PostgreSQL on LinuxONE.

6. Project Deliverables & Production Artifacts

Ready-to-use production software built and verified for end users.

■ **Native Android APK (v1.0.0)** File: releases/macrio-v1.0.0.apk (57 KB) Features instant hardware camera barcode scanning, offline caching, and personalized patient risk profiles. Packaged and zip-aligned.	■ **Progressive Web Application** Live interactive UI on IBM LinuxONE port 8000. Features mobile-responsive glassmorphism dashboard, dynamic Nutri-Grade color tags, and live double-standard comparisons.
■ **Open Source GitHub Repository** URL: https://github.com/manansethia/MAcRIO---Team-TunakTunakTun-IBM-Z-Datathon.git Complete backend, agentic AI scripts, Android source tree, and pre-trained s390x model files.	■ **Machine Learning Model Binaries** File: models/macrio_telum_nutrigrade_ai.joblib (8.9 MB) Offline high-speed Nutri-Grade classifier trained on s390x for rapid offline fallback.

Open Source Attributions: Google Gemma, IBM zDNN, llama.cpp, Open Food Facts, ICMR-NIN IFCT 2017, USDA FoodData Central, FastAPI, PostgreSQL.

7. Impact & Future Roadmap

Delivering consumer empowerment and healthcare savings at enterprise scale.

■ For Consumers & Families Democratizes complex nutritional science. Shoppers simply scan a barcode or point a camera to avoid carcinogens, hidden palm oil, and fraudulent health claims.	**■ For Healthcare Systems** Empowers diabetic and hypertensive patients to proactively prevent emergency hospitalizations and diet-related chronic complications at zero cost.
■■ For Food Regulators Provides automated data-driven evidence of regional ingredient discrimination, equipping civil society and regulators to enforce uniform global quality standards.	**■ Future Enterprise Roadmap** Deploying on IBM Hyper Protect Virtual Servers (HPVS) with confidential container computing, expanding to ASEAN retail supermarkets, and edge IoT scanner integration.

Thank you! Team TunakTunakTun — Proudly running on IBM LinuxONE.